

Exploring Swahili Speech Recognition Model Development

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Abstract

The dominance of a small group of languages on the internet poses a significant challenge to the development of conversational AI tools for low-resource languages. With over 2000 languages spoken in Africa, a third of the world's languages, countries on the continent face a higher proportion of speakers of low-resource languages and lower levels of internet penetration. This is a major concern, especially in access to accurate healthcare information, yet these countries have the highest disease burden. This work focused on fine-tuning a pre-trained XLS-R model, based on wav2vec 2.0 architecture trained with up to 300 million parameters in 128 languages including swahili. With only 3726 transcripts and audios, our fine-tuned model achieved a loss of 1.2834 and a word error rate of 0.5834 on the evaluation set of only 102 sentences. This work was developed on free accessible Google colab, which offers limited computing resources of about 12GB of RAM, and 78GB of disk space. We aim to refine the performance of our baseline model by fine-tuning it first on more general Swahili language data, and subsequently, on healthcare-specific Swahili audio clips and transcripts. In addition, we plan to utilize additional graphical processing resources to further enhance our system's performance. Our ultimate goal is to enable accurate and efficient healthcare documentation in low-resource settings, thereby improving access to quality healthcare for underserved populations.

Keywords: low-resource languages, speech recognition, healthcare documentation, Swahili, underserved populations.

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1. Introduction

There are over 7000 languages in the world. Yet only 10 languages make up for over 70% of global internet users as of January 2023[1], rendering it so difficult to develop conversational AI tools for over 6900 other languages which scramble for less than 30% of the internet usage pie; and they are the low-resource languages. To develop software for all the languages in the world, we would need to go through the painstakingly expensive process of annotating all the languages.

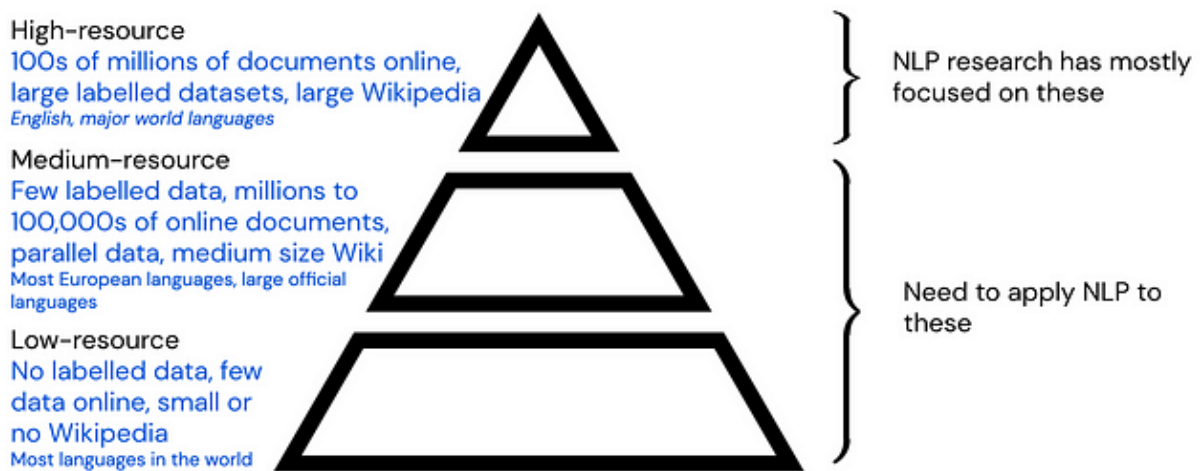


Fig. 1 A conceptual view of the NLP resource hierarchy by Sebastian Ruder[2]

The internet is a major driver for modern information sharing. Limited and close to zero availability of African languages online present many challenges. It is especially a major concern in access to accurate healthcare information, yet these countries have the highest disease burden. The challenge presented in developing conversational tools is not only because there exists fewer data, but there is also a significant challenge with access to computing resources to train machine learning systems for low resources languages. For example, in training the original transformer[3], eight NVIDIA P100 GPUs were used to train their models, with a total of 64 GB of GPU memory. The training process was parallelized across 4 machines, each with 8 GPUs, for a total of 32 GPUs, and training a single model took about 3.5 days. In total, they trained 15 different models, ranging from a "small" model with 6 layers and 512 hidden units to a "big" model with 12 layers and 2048 hidden units. This is infeasible for most resource-constrained setups.

In this work, we aim to push forward the accessibility of conversational tools for healthcare care in swahili speaking communities. We approached this problem by using the Transfer learning technique that leverages the knowledge learned by a model pre-trained on a large dataset. We used swahili data from the Mozilla Common Voice project[4], a publicly available database of crowd-sourced voice datasets for speech recognition software. This was followed by a fine-tuning exercise on the model, Wav2Vec2-XLS-R-300M [5]. We based the

choice of this pre-trained model on three factors. First Wav2Vec2-XLS-R-300M was pre-trained on a large-scale multilingual model for speech. Second, the model architecture is based on wav2vec 2.0.[6] which showed even an hour of labeled fine-tuning data outperformed the previous state-of-the-art on 100 hours subset. And thirdly, the model was trained in 128 languages, including Swahili, making up 436k hours of unlabeled speech.

The fine-tuned model yielded a loss of 1.2834 and a word error rate of 0.5834 on the evaluation set of only 102 transcripts and audios. Subsequently, we conducted more tests on the model using the hosted inference API on Hugging Face. While the model was able to output a readable sentence, it was not without errors. Since the model was a result of fine-tuning on only 3726 transcripts and audio clips, and on the freely accessible version of the Google Colab, we believe that there is still room for improvement. Particularly, we'll focus on fine-tuning the model with more Swahili data and incorporating a language model to improve decoding accuracy. Further, we will fine-tune the model on healthcare-specific Swahili audio clips and transcripts to support patient-doctor consultations hence improving healthcare outcomes.

2. Objectives

The main goal of this project was to develop a swahili speech recognition model that could then be customized to support the documentation of patient-doctor consultations.

3. Methodology

3.1. Data preparation

We used data from common voice to fine-tune a pre-trained model. As of 2019, Common Voice was home to 39 languages with 2,500 hours of collected audio, which has certainly increased by more than 10 folds as of February 2023 as shown in Figure 2.

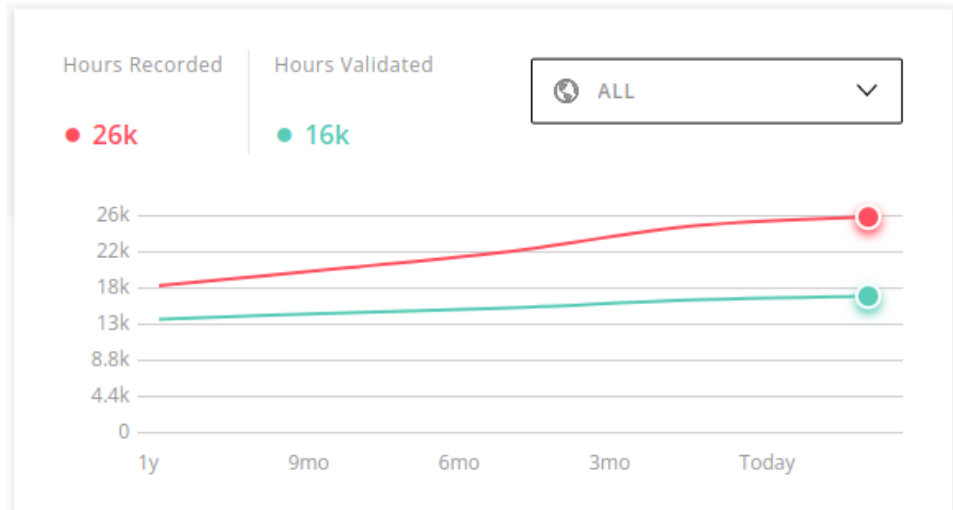


Fig. 2 Statistics of all contributions in common voice

From this corpora, the swahili contained 86,000 sentences and 900 hours of audio with over 300 hours of the audio validated as shown in Figure 3. Considering that low-resource languages are primarily spoken in low-resource regions, our approach to preparing the data and the speech recognition model made use of freely accessible tools to realize a base model and gain an understanding of our approach. Later, we aim to include more data and more graphical computing power to make the model more robust.

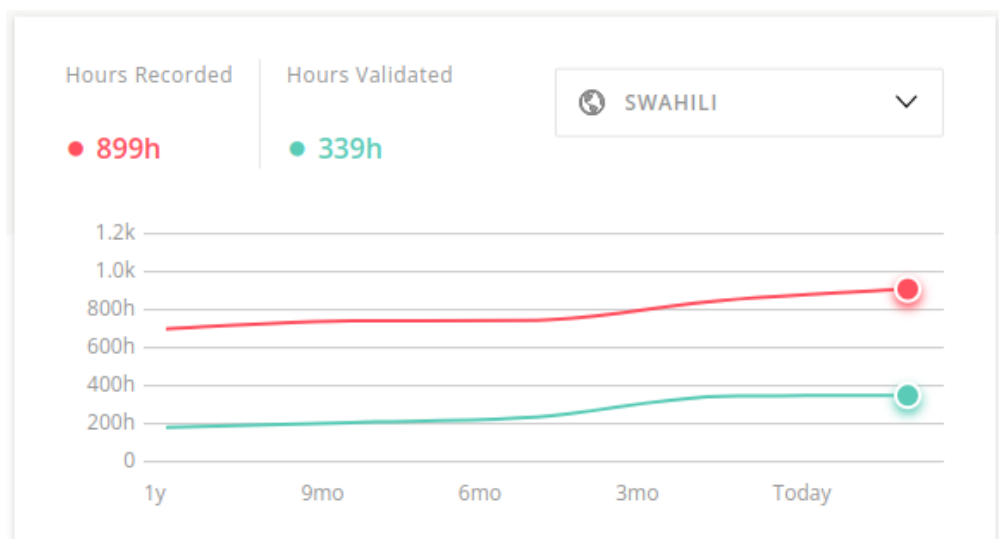


Fig. 3 Swahili statistics on Common Voice

Common Voice data comes with metadata, so we first removed that data and remained with only the audio and its transcript. We then removed punctuation marks and other special characters in the data. Later we converted the audio and clips into an array format for the model to accept them as input. A comprehensive description of the steps of the data preprocessing is published on a blog[7].

3.2. Model development

After completing the preparation of the Swahili clips and transcripts from common voice, we embarked on the fine-tuning exercise. We choose to use a pre-trained model because there exist a good number of pre-trained models that have achieved state-of-the-art performance, especially in natural language tasks, and they are publicly available for free. Wav2Vec2-XLS-R-300M was impressive as it is a large-scale multilingual pre-trained model for speech, based on wav2vec 2.0. Architecture. An architecture that showed even an hour of labeled fine-tuning data outperformed the previous state-of-the-art on 100 hours subset. Additionally, the Wav2Vec2-XLS-R-300M model was trained in 128 languages, including Swahili, making up 436k hours of unlabeled speech.

We added a CTC loss[8] on top of the transformer to perform the speech recognition task as illustrated in Figure 4. A description of how the steps we took to fine-tune can are documented on a blog[9] and the code is hosted on GitHub.

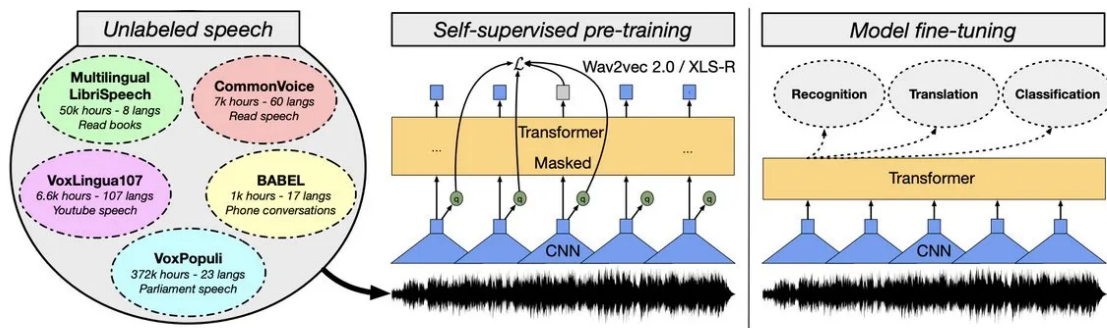


Figure 1: **Self-supervised cross-lingual representation learning.** We pre-train a large multilingual wav2vec 2.0 Transformer (XLS-R) on 436K hours of unannotated speech data in 128 languages. The training data is from different public speech corpora and we fine-tune the resulting model for several multilingual speech tasks.

Fig. 4 XLS-R: Self-supervised Cross-Lingual Speech Representation Learning at Scale

4. Results and Discussion

The evaluation of the Swahili speech recognition model yielded a loss of 1.2834 and a word error rate of 0.5834 on the evaluation set of just 102 sentences. Subsequently, we conducted more tests on the model using the hosted inference API on Hugging Face[10]. While the model was able to output a readable sentence, it was not without errors. For example, the output in Figure 5 is a result of the model's output. The ground truth sentence is “walinzi wachezaji wa kiungo na wachezaji wa mbele au strika.” The model makes some errors. Notably, instead of “walinzi” it breaks the word into “walim zi” and inserts a “u” on the word “strika”.

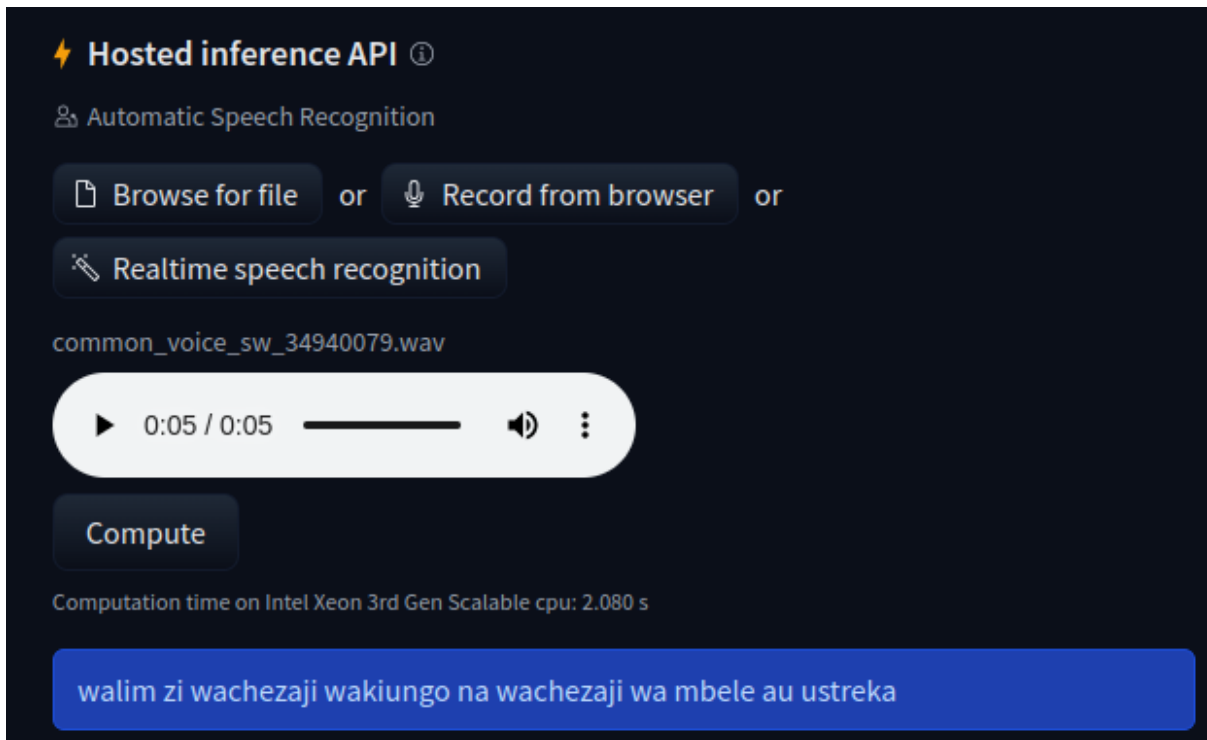


Fig. 5 Testing the fine-tuned model on Hugging face inference API

This is a somewhat readable sentence and considering we just fine-tuned the model on only 3726 sentences, it was well done. It also gives us confidence that by adding the number of data, and adjusting the hyperparameters accordingly e.g adding more epochs from 9 to a higher number, while making use of a more reliable GPU-powered machine, we will have a high-performing model. Other ways we are exploiting include adding another layer on top of the CTC loss function to improve the decoding accuracy of our model.

Our objective for this model however was to develop a system that can support patient-doctor consultation documentation. The results of this work thus far are a breakthrough toward realizing that goal. However, we still need to get patient-doctor conversations data and then inject them into our model for training then test our model on different conversations that it

has not seen. Achieving acceptable performance will be followed by putting the model in the hands of the users – doctors and their patients followed by robust evaluation.

5. Future work

Although the model performed well overall given that we only fine-tuned it with 3726 transcripts and audios, the errors we detailed in the discussion indicates that there is still room for improvement. The immediate action is to add more data and adjust the hyperparameters of the model accordingly, then use a stronger GPU system to train a more robust model. That will be followed by further fine-tuning on a healthcare dataset to train the model to transcribe patient-doctor conversations.

6. Conclusion

We have successfully developed a Swahili speech recognition model by fine-tuning the Wav2Vec2-XLS-R-300M model and publicly hosted it on Hugging Face Hub. The model was is a result of fine-tuning the Wav2Vec2-XLS-R-300M model using 3726 swahili transcripts and audios, achieving a loss of 1.2834 and a word error rate of 0.5834 on the evaluation set. The entire model development took place on a Google Colab notebook, the freely available version which offers limited graphical computing power of ~ 12 GB of RAM, and ~ 78 GB of disk space. This work presents preliminary results and initial milestones toward achieving a robust model that can transcribe patient-doctor consultations.

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